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Exact outage analysis of underlay cooperative cognitive networks over Nakagami-*m* fading channels

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Abstract: This paper studies the outage performance behaviour of underlay cooperative cognitive networks over Nakagami-*m* fading channels. For time savings in collecting simulation results, the authors propose a new outage probability expression. Notably, this expression is quite general, taking into account practical operation conditions such as interference power constraint, maximum transmit power constraint, correlation among received signal-to-noise ratios, non-identically and independently distributed Nakagami-*m* fading channels and the maximum ratio combining receiver. Moreover, it is useful in obtaining numerical results for performance comparison between dual-hop communications and cooperative communications in underlay cognitive networks. Interestingly, analytical results illustrate that like dual-hop cognitive networks, cooperative cognitive ones experience an error floor but the latter significantly outperform the former at almost no cost of power and bandwidth.

1 Introduction

The conflict in spectrum resource utilisation nowadays becomes serious. On the one hand, the spectrum utilisation efficiency of licensed users is very low [1]. On the other hand, the available spectrum for new wireless applications is very limited [2]. Cognitive radio technology aims to effectively solve this conflict [3]. Indeed, unlicensed users (or cognitive radios) with the capability of controlling the interference level at licensed users can efficiently utilise the frequency band allotted to licensed users without harming the communication of licensed users. Popularly, cognitive radios operate in the underlay mode where their transmit power is intelligently turned such that the interference at licensed users is always below an allowable level [4–6].

It is the transmit power constraint that limits the transmission range of unlicensed users. To overcome this weakness of cognitive radio networks, the cognitive radio technology should combine cooperative communications [Multi-hop communications (i.e. [7–9]) does not take the advantage of the direct channel between the source and the destination whereas cooperative communications does for enhanced spatial diversity (i.e. [10–14]).] techniques, forming underlay cooperative cognitive networks. Indeed, with the advantage of space diversity, cooperative communications can extend the coverage without the need of physical antenna arrays [15]. In most cooperative communication protocols, the helpers (or the relays) can process received signals in either amplify-and-forward (AF) or decode-and-forward (DF) manner [16].

Outage probability analysis is very important to get a useful insight into the information-theoretic performance limit [17]. It is our objective in this paper. Before elaborately discussing some related works on the outage probability analysis of underlay DF-type cooperative cognitive networks in [10–14], it is reminded that underlay cooperative cognitive networks reveal some notable characteristics such as the transmit power constraint on unlicensed users, the correlation among received SNRs, fading distribution, and receiver structure. More specifically, which characteristics included in existing works are summarised in Table 1:

- *About transmit power constraint:* In [10, 11, 14] the authors study the outage performance under only the interference power constraint whereas [12, 13] solve more general problem with investigation of both interference power constraint and maximum transmit power constraint.
- *About the correlation among received SNRs*: In [11–14] the authors consider this correlation in the analysis to evaluate the system performance more accurately than [10] that assumes independent SNRs will be included.
- *About fading distribution*: In [10–13] the authors assume Rayleigh fading. Nevertheless, among them only [13] investigates non-identically and independently distributed (i.i.d.) channels. It is well known that most fading channels can be statistically characterised with the Nakagami-*m* distribution, including the Rayleigh distribution ($m = 1$) as a special case [18]. Therefore it is essential to analyse the performance of communication systems over Nakagami-*m* fading channels. However, only Duong *et al.* [14] has recently mentioned non-i.i.d. Nakagami-*m* fading channels [Guo *et al.* [10] also studied non-i.i.d. Nakagami-*m* fading channels but for 'dual-hop' cognitive networks. Thus, the correlation among received SNRs is neglected.].

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Table 1 Summary on characteristics investigated in [10–14]

Ref.	Power cons.		Corr.	Fading				Receiver	
	Int.	Trans.		Nak.	Ray.	Non-i.i.d.	i.i.d.	MRC	SC
[10]	x	—	—	—	x	—	x	x	—
[11]	x	—	x	—	x	—	x	x	—
[12]	x	x	x	—	x	—	x	x	—
[13]	x	x	x	—	x	x	x	x	—
[14]	x	—	x	x	x	x	x	—	x

Abbreviations used: 'Ref.' for 'Reference', 'Power cons.' for 'Transmit power constraint', 'Fading' for 'Fading distribution', 'Int.' for 'Interference power constraint', 'Trans.' for 'Maximum transmit power constraint', 'Corr.' for 'Correlation among received SNRs', 'Nak.' for 'Nakagami-*m*', 'Ray.' for 'Rayleigh'

• *About the receiver structure*: Only Duong *et al.* [14] investigates the selection combining (SC) receiver whereas the others, [10–13], study the maximum ratio combining (MRC) receiver. It is noted that the MRC receiver is better than the SC receiver [18]. Thus, for performance limit investigation, the MRC receiver deserves to be studied.

From the above literature review, a question raised is that how the MRC receiver in underlay DF-type cooperative cognitive networks performs under practical conditions such as interference power constraint, maximum transmit power constraint, the correlation among received SNRs, and non-i.i.d. Nakagami-*m* fading. We will answer this question. For time savings in collecting simulation results, we propose a new outage probability expression. This expression benefits not only the performance evaluation of underlay cooperative cognitive networks in key operation parameters but also performance comparison between cooperative and dual-hop cognitive networks. Since the cooperative and dual-hop communications consume almost the same system resources such as power and bandwidth, the answer on how much performance gain the former can achieve over the latter is very important. We will provide a variety of analytical results from which we can have insights into system performance behaviour as follows: (i) Like dual-hop cognitive networks, cooperative cognitive ones suffer the performance saturation phenomena at either high maximum interference power or high maximum transmit power; (ii) The cooperative cognitive networks dramatically outperform the dual-hop cognitive ones at almost no cost of power and bandwidth.

We organise this paper as follows. Section 2 presents the system model. The detailed outage probability analysis is discussed in Section 3. Analytical and simulated results are provided in Section 4 for performance assessment and comparison between the dual-hop communications and the cooperative communications in underlay cognitive networks. We close the paper with conclusions in Section 5.

2 System model

We consider a typical underlay DF-type cooperative cognitive network in Fig. 1. Two-stage cooperative communications is performed in the unlicensed network in which the #1 user assists information transmission from the #0 user to the #2 user, and the activation of unlicensed users interferes the operation of the licensed user, namely the #3 user. In the stage 1, the #0 user sends its information and in the stage 2, the #1 user is either activated if successfully decoding the #0 user's information or idle if otherwise. Upon the #1 user's decoding status, the #2 user performs the restoration of the #0 user's information with either MRC both signals from the #0 user and the #1 user, or based on only the signal from the #0 user. It is noted that, this communication model is reduced to the dual-hop communications in [9] by ignoring the channel between the #0 user and the #2 user.

We assume non-i.i.d. frequency-flat Nakagami-*m* fading channels. Then, the probability density function (pdf) of the channel power $p_{ab} = |c_{ab}|^2$, where c_{ab} is the channel coefficient between the #*a* transmitter and the #*b* receiver, is given by

$$f_{p_{ab}}(x) = \frac{\beta_{ab}^{m_{ab}}}{\Gamma(m_{ab})} x^{m_{ab}-1} \mathrm{e}^{-\beta_{ab}x} \tag{1}$$

where $\beta_{ab} = m_{ab}/\lambda_{ab}$ with m_{ab} and λ_{ab} being the fading severity parameter and the average fading power, respectively; $\Gamma(\cdot)$ is the gamma function [19]; $a \in \{0, 1\}$; $b \in \{1, 2, 3\}$. Owing to non-i.i.d. fading, m_{ab}'s are not necessarily identical (similarly to λ_{ab}'s).

Let T_a be the transmit power of the #*a* unlicensed user with $a \in \{0, 1\}$. In underlay cognitive networks (e.g. [7]), the setting of $T_a = \min((R_m/p_{a3}), T_m)$ is required to satisfy both interference power constraint, $T_a \le (R_m/p_{a3})$, and maximum transmit power constraint, $T_a \le T_m$ as well as to obtain the maximum transmission range. Here, R_m is the maximum interference power that the licensed user can tolerate, and T_m is the maximum transmit power that the unlicensed user can have.

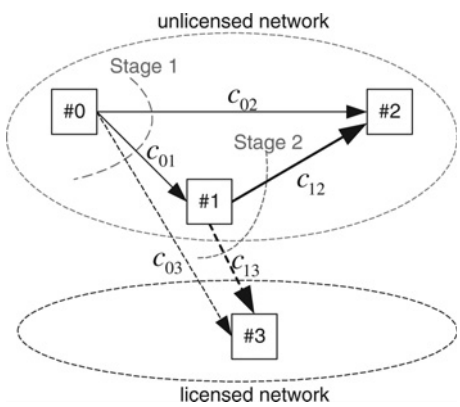

Fig. 1 *System model*

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The channel between the #a transmitter and the #b receiver results in the received SNR as [9]

$$s_{ab} = T_a \frac{p_{ab}}{N_0} = \min\left(\frac{R_m}{p_{a3}}, T_m\right) \frac{p_{ab}}{N_0} \tag{2}$$

where N_0 is the normalised noise variance at the receivers.

The correlation between received SNRs can be interpreted as follows. From (2), the received SNRs at the #1 user and the #2 user in the stage 1 are $s_{01} = \min\big((R_m/p_{03}), T_m\big)(p_{01}/N_0)$ and $s_{02} = \min\big((R_m/p_{03}), T_m\big)(p_{02}/N_0)$, respectively. Since both s_{01} and s_{02} contain p_{03}, they are correlated.

We integrate all characteristics of underlay cooperative cognitive networks such as maximum transmit power constraint, interference power constraint, the correlation among received SNRs, non-i.i.d. Nakagami-m fading and the MRC receiver into our system model. As a result, our analysis is more complicated and general than the existing works in [10–14].

3 Performance analysis

Owing to the two-stage nature of the cooperative communication protocol under investigation, the receiver is in outage if the received SNR, s, satisfies the inequality $G > (1/2)\log_2(1+s)$ or $s < k$ with $k = 2^{2G}-1$ and G being the transmission rate of the unlicensed network. Based on the operation principle of this protocol, the #2 user is in outage either if both #1 user and #2 user are in outage (i.e. $s_{01} < k$ and $s_{02} < k$) or if only the #2 user is in outage (i.e. $s_{01} \geq k$ and $s_{02} + s_{12} < k$). In summary, the outage probability of underlay cooperative cognitive networks is given as

$$\mathrm{OP} = \underbrace{\Pr\{s_{02} < k,\ s_{01} < k\}}_{\mathcal{I}_1} + \underbrace{\Pr\{s_{02} + s_{12} < k,\ s_{01} \geq k\}}_{\mathcal{I}_2} \tag{3}$$

where $\Pr\{x\}$ is the probability of the event x.

The exact closed forms of $\mathcal{I}_1$ and $\mathcal{I}_2$ in (3) are provided in next subsections.

3.1 Derivation of $\mathcal{I}_1$

Given s_{01} and s_{02} in the compact form in (2), we rewrite $\mathcal{I}_1$ as (see (4))

$$\begin{aligned}\mathcal{I}_1 &= \Pr\left\{p_{02} < \frac{N_0 k}{\min\big((R_m/p_{03}),\ T_m\big)},\ p_{01} < \frac{N_0 k}{\min\big((R_m/p_{03}),\ T_m\big)}\right\} \\ &= \int_0^\infty \Pr\left\{p_{02} < \frac{N_0 k}{\min\big((R_m/x),\ T_m\big)}\right\} \Pr\left\{p_{01} < \frac{N_0 k}{\min\big((R_m/x),\ T_m\big)}\right\} f_{p_{03}}(x)\ \mathrm{d}x\end{aligned} \tag{4}$$

where $f_{p_{03}}(x)$ is the pdf of p_{03}.

Given the distribution of p_{ab} in (1), we have the following results:

- *Result (A):* $\Pr\{p_{ab} < x\} = \dfrac{\gamma(m_{ab},\ \beta_{ab}x)}{\Gamma(m_{ab})}$.
- *Result (B):* $\Pr\{p_{ab} > x\} = \dfrac{\Gamma(m_{ab},\ \beta_{ab}x)}{\Gamma(m_{ab})}$.

where

$$\gamma(a, x) = \int_0^x \mathrm{e}^{-t} t^{a-1}\mathrm{d}t \text{ and } \Gamma(a, x) = \int_x^\infty \mathrm{e}^{-t} t^{a-1}\mathrm{d}t$$

are the 'lower' and 'upper' incomplete gamma functions in [19, eq. (8.350.1)] and [19, eq. (8.350.2)], respectively.

Using the result ($\mathcal{A}$) reduces $\mathcal{I}_1$ to (see (5))

$$\begin{aligned}\mathcal{I}_1 &= \int_0^\infty \frac{\gamma\big(m_{02},\ ((\beta_{02}N_0k)/(\min((R_m/x),\ T_m)))\big)}{\Gamma(m_{02})} \frac{\gamma\big(m_{01},\ ((\beta_{01}N_0k)/(\min((R_m/x),\ T_m)))\big)}{\Gamma(m_{01})} f_{p_{03}}(x)\ \mathrm{d}x \\ &= \underbrace{\int_\mu^\infty \big((\gamma(m_{02},\ \beta_{02}vkx))/(\Gamma(m_{02}))\big)\big((\gamma(m_{01},\ \beta_{01}vkx))/(\Gamma(m_{01}))\big) f_{p_{03}}(x)\ \mathrm{d}x}_{\mathcal{I}_{11}} \\ &\quad + \underbrace{\int_0^\mu \big((\gamma(m_{02},\ \beta_{02}\chi k))/(\Gamma(m_{02}))\big)\big((\gamma(m_{01},\ \beta_{01}\chi k))/(\Gamma(m_{01}))\big) f_{p_{03}}(x)\ \mathrm{d}x}_{\mathcal{I}_{12}}\end{aligned} \tag{5}$$

where constants $\{\mu, \chi, v\}$ are defined as

$$\mu = \frac{R_m}{T_m} \tag{6}$$

$$\chi = \frac{N_0}{T_m} \tag{7}$$

$$v = \frac{N_0}{R_m} \tag{8}$$

Now, the derivation of $\mathcal{I}_1$ is completed after evaluating $\mathcal{I}_{11}$ and $\mathcal{I}_{12}$. First of all, the closed form of $\mathcal{I}_{12}$ is obtained with the help of the result ($\mathcal{A}$)

$$\begin{aligned}\mathcal{I}_{12} &= \frac{\gamma(m_{01},\ \beta_{01}\chi k)\gamma(m_{02},\ \beta_{02}\chi k)}{\Gamma(m_{01})\Gamma(m_{02})} \int_0^\mu f_{p_{03}}(x)\ \mathrm{d}x \\ &= \frac{\gamma(m_{01},\ \beta_{01}\chi k)\gamma(m_{02},\ \beta_{02}\chi k)\gamma(m_{03},\ \beta_{03}\mu)}{\Gamma(m_{01})\Gamma(m_{02})\Gamma(m_{03})}\end{aligned} \tag{9}$$

To derive the closed form of $\mathcal{I}_{11}$, it is recalled from [9, eq.

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(22)] and [19, eq. (8.352.1)] that

$$\gamma(n, x) = \Gamma(n)\left(1 - \mathrm{e}^{-x}\sum_{i=0}^{n-1}\frac{x^i}{i!}\right) \tag{10}$$

Applying (10) together with (1), we obtain the closed form of $\mathcal{I}_{11}$ as (see (11))

The integrals in (11) are solved in closed-form based on the definition of $\Gamma(a, x)$.

3.2 Derivation of $\mathcal{I}_2$

With s_{01} and s_{02} given in the general form in (2), $\mathcal{I}_2$ is rewritten as (see (12))

where $f_{s_{12}}(y)$ is the pdf of s_{12}. To find its closed form, it is reminded that the cumulative density function (cdf) of s_{12} is given in [9, eq. (14)] as

$$F_{s_{12}}(x) = \frac{\gamma(m_{12},\ \beta_{12}\chi x)}{\Gamma(m_{12})}\frac{\gamma(m_{13},\ \beta_{13}\mu)}{\Gamma(m_{13})} + \frac{\Gamma(m_{13},\ \beta_{13}\mu)}{\Gamma(m_{13})} - \frac{\beta_{13}^{m_{13}}}{\Gamma(m_{13})}\sum_{i=0}^{m_{12}-1}\frac{(\beta_{12}v)^i}{i!}\frac{x^i\Gamma(m_{13}+i,\ \beta_{13}\mu+\beta_{12}\chi x)}{(\beta_{13}+\beta_{12}vx)^{m_{13}+i}} \tag{13}$$

Taking the derivative of $F_{s_{12}}(x)$ with respect to x results in the closed form of $f_{s_{12}}(x)$ as

$$\begin{aligned} f_{s_{12}}(x) = {} & \frac{(\beta_{12}\chi)^{m_{12}}\gamma(m_{13},\ \beta_{13}\mu)}{\Gamma(m_{13})\Gamma(m_{12})}x^{m_{12}-1}\mathrm{e}^{-\beta_{12}\chi x} \\ & - \frac{\beta_{13}^{m_{13}}}{\Gamma(m_{13})}\sum_{i=1}^{m_{12}-1}\frac{(\beta_{12}v)^i}{(i-1)!} \\ & \times \frac{x^{i-1}\Gamma(m_{13}+i,\ \beta_{13}\mu+\beta_{12}\chi x)}{(\beta_{13}+\beta_{12}vx)^{m_{13}+i}} \\ & + \frac{\beta_{13}^{m_{13}}\mathrm{e}^{-\beta_{13}\mu}}{\Gamma(m_{13})}\sum_{i=0}^{m_{12}-1}\frac{\beta_{12}^{i+1}v^i\chi}{i!\mu^{-m_{13}-i+1}}\frac{x^i\mathrm{e}^{-\beta_{12}\chi x}}{\beta_{13}+\beta_{12}vx} \\ & + \frac{\beta_{13}^{m_{13}}}{\Gamma(m_{13})}\sum_{i=0}^{m_{12}-1}\frac{(m_{13}+i)}{i!(\beta_{12}v)^{-i-1}} \\ & \times \frac{x^i\Gamma(m_{13}+i,\ \beta_{13}\mu+\beta_{12}\chi x)}{(\beta_{13}+\beta_{12}vx)^{m_{13}+i+1}} \end{aligned} \tag{14}$$

$$\begin{aligned} \mathcal{I}_{11} &= \int_\mu^\infty \left(1 - \mathrm{e}^{-\beta_{02}vkx}\sum_{i=0}^{m_{02}-1}\frac{(\beta_{02}vkx)^i}{i!}\right)\left(1 - \mathrm{e}^{-\beta_{01}vkx}\sum_{j=0}^{m_{01}-1}\frac{(\beta_{01}vkx)^j}{j!}\right)\frac{\beta_{03}^{m_{03}}}{\Gamma(m_{03})}x^{m_{03}-1}\mathrm{e}^{-\beta_{03}x}\mathrm{d}x \\ &= \int_\mu^\infty \frac{\beta_{03}^{m_{03}}}{\Gamma(m_{03})}x^{m_{03}-1}\mathrm{e}^{-\beta_{03}x}\mathrm{d}x - \frac{\beta_{03}^{m_{03}}}{\Gamma(m_{03})}\sum_{i=0}^{m_{02}-1}\frac{(\beta_{02}vk)^i}{i!}\int_\mu^\infty x^{i+m_{03}-1}\mathrm{e}^{-(\beta_{02}vk+\beta_{03})x}\mathrm{d}x \\ &\quad - \frac{\beta_{03}^{m_{03}}}{\Gamma(m_{03})}\sum_{j=0}^{m_{01}-1}\frac{(\beta_{01}vk)^j}{j!}\int_\mu^\infty x^{i+m_{03}-1}\mathrm{e}^{-(\beta_{01}vk+\beta_{03})x}\mathrm{d}x \\ &\quad + \frac{\beta_{03}^{m_{03}}}{\Gamma(m_{03})}\sum_{i=0}^{m_{02}-1}\sum_{j=0}^{m_{01}-1}\frac{\beta_{02}^i\beta_{01}^j(vk)^{i+j}}{i!j!}\int_\mu^\infty x^{i+j+m_{03}-1}\mathrm{e}^{-(\{\beta_{02}+\beta_{01}\}vk+\beta_{03})x}\mathrm{d}x \\ &= \frac{\Gamma(m_{03},\ \beta_{03}\mu)}{\Gamma(m_{03})} - \frac{\beta_{03}^{m_{03}}}{\Gamma(m_{03})}\sum_{i=0}^{m_{02}-1}\frac{(\beta_{02}vk)^i\Gamma(i+m_{03},\ (\beta_{02}vk+\beta_{03})\mu)}{i!(\beta_{02}vk+\beta_{03})^{i+m_{03}}} \\ &\quad - \frac{\beta_{03}^{m_{03}}}{\Gamma(m_{03})}\sum_{j=0}^{m_{01}-1}\frac{(\beta_{01}vk)^j\Gamma(j+m_{03},\ (\beta_{01}vk+\beta_{03})\mu)}{j!(\beta_{01}vk+\beta_{03})^{j+m_{03}}} \\ &\quad + \frac{\beta_{03}^{m_{03}}}{\Gamma(m_{03})}\sum_{i=0}^{m_{02}-1}\sum_{j=0}^{m_{01}-1}\frac{\beta_{02}^i\beta_{01}^j(vk)^{i+j}\Gamma(i+j+m_{03},\ (\{\beta_{02}+\beta_{01}\}vk+\beta_{03})\mu)}{i!j!(\{\beta_{02}+\beta_{01}\}vk+\beta_{03})^{i+j+m_{03}}} \end{aligned} \tag{11}$$

$$\begin{aligned} \mathcal{I}_2 &= \Pr\left\{\min\left(\frac{R_m}{p_{03}},\ T_m\right)\frac{p_{02}}{N_0} + s_{12} < k,\quad \min\left(\frac{R_m}{p_{03}},\ T_m\right)\frac{p_{01}}{N_0} \geq k\right\} \\ &= \int_0^k\left(\int_0^\infty \Pr\left\{p_{02} < \frac{(k-y)N_0}{\min(((R_m)/x),\ T_m)},\quad p_{01} \geq \frac{N_0k}{\min(((R_m)/x),\ T_m)}\right\}f_{p_{03}}(x)\ \mathrm{d}x\right)f_{s_{12}}(y)\ \mathrm{d}y \\ &= \int_0^k\left(\int_0^\infty \Pr\left\{p_{02} < \frac{(k-y)N_0}{\min(((R_m)/x),\ T_m)}\right\}\Pr\left\{p_{01} \geq \frac{N_0k}{\min(((R_m)/x),\ T_m)}\right\}f_{p_{03}}(x)\ \mathrm{d}x\right)f_{s_{12}}(y)\ \mathrm{d}y \end{aligned} \tag{12}$$

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Applying the results ($\mathcal{A}$) and ($\mathcal{B}$), we reduce $\mathcal{I}_2$ to (see (15))

The problem now is to complete the computation of $\mathcal{I}_{21}$ and $\mathcal{I}_{22}$. First of all, it is recalled from [19, eq. (8.352.2)] that

$$\Gamma(n, x) = \Gamma(n)\mathrm{e}^{-x}\sum_{i=0}^{n-1}\frac{x^i}{i!} \tag{16}$$

Using (10) and (16) together with (1), the inner integral of $\mathcal{I}_{21}$ is rewritten as (see (17))

Plugging (17) in (15), $\mathcal{I}_{21}$ is reduced to (see (18))

It is obvious that

$$\mathcal{I}_{211} = F_{s_{12}}(k) \tag{19}$$

Inserting (14) into $\mathcal{I}_{212}$, we obtain

$$\begin{aligned}\mathcal{I}_{212} = {} & \frac{(\beta_{12}\chi)^{m_{12}}\gamma(m_{13},\ \beta_{13}\mu)}{\Gamma(m_{13})\Gamma(m_{12})} f_{212b}^{m_{12}-1,m_{03}+j+s,0} \\ & - \frac{\beta_{13}^{m_{13}}}{\Gamma(m_{13})}\sum_{i=1}^{m_{12}-1}\frac{(\beta_{12}v)^i}{(i-1)!} f_{212a}^{i-1,m_{03}+j+s,m_{13}+i,m_{13}+i} \\ & + \frac{\beta_{13}^{m_{13}}\mathrm{e}^{-\beta_{13}\mu}}{\Gamma(m_{13})}\sum_{i=0}^{m_{12}-1}\frac{\beta_{12}^{i+1}v^i\chi}{i!\mu^{-m_{13}-i+1}} f_{212b}^{i,m_{03}+j+s,1} \\ & + \frac{\beta_{13}^{m_{13}}}{\Gamma(m_{13})}\sum_{i=0}^{m_{12}-1}\frac{(m_{13}+i)}{i!(\beta_{12}v)^{-i-1}} f_{212a}^{i,m_{03}+j+s,m_{13}+i,m_{13}+i+1}\end{aligned} \tag{20}$$

$$\begin{aligned}\mathcal{I}_2 &= \int_0^k\left[\int_0^\infty \frac{\gamma(m_{02},\ ((\beta_{02}(k-y)N_0)/(\min((R_m/x),\ T_m))))}{\Gamma(m_{02})}\frac{\Gamma(m_{01},\ ((\beta_{01}N_0k)/(\min((R_m/x),\ T_m))))}{\Gamma(m_{01})} f_{p_{03}}(x)\ \mathrm{d}x\right] f_{s_{12}}(y)\ \mathrm{d}y \\ &= \underbrace{\int_0^k \underbrace{\left\{\int_\mu^\infty \frac{\gamma(m_{02},\ \beta_{02}v(k-y)x)}{\Gamma(m_{02})}\frac{\Gamma(m_{01},\ \beta_{01}vkx)}{\Gamma(m_{01})} f_{p_{03}}(x)\ \mathrm{d}x\right\}}_{\mathcal{I}_{21i}} f_{s_{12}}(y)\ \mathrm{d}y}_{\mathcal{I}_{21}} \\ &\quad + \frac{\Gamma(m_{01},\ \beta_{01}\chi k)\gamma(m_{03}, \beta_{03}\mu)}{\Gamma(m_{01})\Gamma(m_{03})}\underbrace{\int_0^k \frac{\gamma(m_{02},\ \beta_{02}\chi(k-y))}{\Gamma(m_{02})} f_{s_{12}}(y)\ \mathrm{d}y}_{\mathcal{I}_{22}}\end{aligned} \tag{15}$$

$$\begin{aligned}\mathcal{I}_{21i} &= \int_\mu^\infty\left(1-\mathrm{e}^{-\beta_{02}v(k-y)x}\sum_{j=0}^{m_{02}-1}\frac{[\beta_{02}v(k-y)x]^j}{j!}\right)\left(\mathrm{e}^{-\beta_{01}vkx}\sum_{s=0}^{m_{01}-1}\frac{(\beta_{01}vkx)^s}{s!}\right)\frac{\beta_{03}^{m_{03}}}{\Gamma(m_{03})}x^{m_{03}-1}\mathrm{e}^{-\beta_{03}x}\mathrm{d}x \\ &= \frac{\beta_{03}^{m_{03}}}{\Gamma(m_{03})}\sum_{s=0}^{m_{01}-1}\frac{(\beta_{01}vk)^s}{s!}\int_\mu^\infty x^{m_{03}+s-1}\mathrm{e}^{-(\beta_{01}vk+\beta_{03})x}\mathrm{d}x \\ &\quad - \frac{\beta_{03}^{m_{03}}}{\Gamma(m_{03})}\sum_{j=0}^{m_{02}-1}\sum_{s=0}^{m_{01}-1}\frac{[\beta_{02}v(k-y)]^j(\beta_{01}vk)^s}{j!s!}\int_\mu^\infty x^{m_{03}+j+s-1}\mathrm{e}^{-(\beta_{02}v(k-y)+\beta_{01}vk+\beta_{03})x}\mathrm{d}x \\ &= \frac{\beta_{03}^{m_{03}}}{\Gamma(m_{03})}\sum_{s=0}^{m_{01}-1}\frac{(\beta_{01}vk)^s\Gamma(m_{03}+s,\ (\beta_{01}vk+\beta_{03})\mu)}{s!(\beta_{01}vk+\beta_{03})^{m_{03}+s}} \\ &\quad - \frac{\beta_{03}^{m_{03}}}{\Gamma(m_{03})}\sum_{j=0}^{m_{02}-1}\sum_{s=0}^{m_{01}-1}\frac{[\beta_{02}v(k-y)]^j}{j!s!(\beta_{01}vk)^{-s}}\frac{\Gamma(m_{03}+j+s,\ (\beta_{02}v(k-y)+\beta_{01}vk+\beta_{03})\mu)}{(\beta_{02}v(k-y)+\beta_{01}vk+\beta_{03})^{m_{03}+j+s}}\end{aligned} \tag{17}$$

$$\begin{aligned}\mathcal{I}_{21} &= \frac{\beta_{03}^{m_{03}}}{\Gamma(m_{03})}\sum_{s=0}^{m_{01}-1}\frac{(\beta_{01}vk)^s\Gamma(m_{03}+s,\ (\beta_{01}vk+\beta_{03})\mu)}{s!(\beta_{01}vk+\beta_{03})^{m_{03}+s}}\underbrace{\int_0^k f_{s_{12}}(y)\ \mathrm{d}y}_{\mathcal{I}_{211}} \\ &\quad - \frac{\beta_{03}^{m_{03}}}{\Gamma(m_{03})}\sum_{j=0}^{m_{02}-1}\sum_{s=0}^{m_{01}-1}\frac{(\beta_{02}v)^j}{j!s!(\beta_{01}vk)^{-s}}\underbrace{\int_0^k\frac{\Gamma(m_{03}+j+s,\ (\beta_{02}v(k-y)+\beta_{01}vk+\beta_{03})\mu)}{(k-y)^{-j}(\beta_{02}v(k-y)+\beta_{01}vk+\beta_{03})^{m_{03}+j+s}} f_{s_{12}}(y)\ \mathrm{d}y}_{\mathcal{I}_{212}}\end{aligned} \tag{18}$$

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In (20), we define two integrals (see (21) and (22))

These two integrals are expressed in the closed form in (41) and (42), respectively, in the appendix. Putting (19) and (20) in (18), we obtain the closed form of $\mathcal{I}_{21}$.

Now we move to compute $\mathcal{I}_{22}$ for completing (15). Applying (10) and the binomial expansion $(a+b)^n = \sum_{i=0}^{n} \binom{n}{i} a^i b^{n-i}$ with $\binom{a}{b} = \frac{a!}{b!(a-b)!}$ sequentially, the closed form of $\mathcal{I}_{22}$ is given as

$$\mathcal{I}_{22} = \int_0^k \left(1 - \mathrm{e}^{-\beta_{02}\chi(k-y)} \sum_{w=0}^{m_{02}-1} \frac{\left[\beta_{02}\chi(k-y)\right]^w}{w!}\right) f_{s_{12}}(y)\,\mathrm{d}y$$
$$= F_{s_{12}}(k) - \mathrm{e}^{-\beta_{02}\chi k} \sum_{w=0}^{m_{02}-1} \sum_{u=0}^{w} \binom{w}{u} \frac{(-1)^u k^{w-u} (\beta_{02}\chi)^w}{w!}$$
$$\underbrace{\int_0^k \mathrm{e}^{\beta_{02}\chi y} y^u f_{s_{12}}(y)\,\mathrm{d}y}_{\mathcal{I}_{221}} \tag{23}$$

Plugging (14) in $\mathcal{I}_{221}$, we deduce its closed form as

$$\mathcal{I}_{221} = \frac{(\beta_{12}\chi)^{m_{12}} \gamma(m_{13},\ \beta_{13}\mu)}{\Gamma(m_{13})\Gamma(m_{12})} f_{221a}^{m_{12}+u-1,(\beta_{02}-\beta_{12})\chi;0,k}$$
$$- \frac{\beta_{13}^{m_{13}}}{\Gamma(m_{13})} \sum_{i=1}^{m_{12}-1} \frac{(\beta_{12}v)^i}{(i-1)!} f_{221c}^{i+u-1,m_{13}+i,m_{13}+i}$$
$$+ \frac{\beta_{13}^{m_{13}} \mathrm{e}^{-\beta_{13}\mu}}{\Gamma(m_{13})} \sum_{i=0}^{m_{12}-1} \frac{\beta_{12}^{i+1} v^i \chi \mu^{m_{13}+i-1}}{i!} f_{221b}^{i+u,1}$$
$$+ \frac{\beta_{13}^{m_{13}}}{\Gamma(m_{13})} \sum_{i=0}^{m_{12}-1} \frac{(m_{13}+i)(\beta_{12}v)^{i+1}}{i!} f_{221c}^{i+u,m_{13}+i,m_{13}+i+1} \tag{24}$$

In (24), three integrals are defined as

$$f_{221a}^{n,\beta;\Omega,\Lambda} = \int_{\Omega}^{\Lambda} x^n \mathrm{e}^{\beta x}\mathrm{d}x \tag{25}$$

$$f_{221b}^{a,b} = \int_0^k \frac{x^a \mathrm{e}^{(\beta_{02}-\beta_{12})\chi x}}{(\beta_{13}+\beta_{12}vx)^b}\mathrm{d}x \tag{26}$$

$$f_{221c}^{a,b,c} = \int_0^k \frac{x^a \Gamma(b,\ \beta_{13}\mu + \beta_{12}\chi x)\mathrm{e}^{\beta_{02}\chi x}}{(\beta_{13}+\beta_{12}vx)^c}\mathrm{d}x \tag{27}$$

All of them are expressed in the closed form as (28)–(33), correspondingly in the appendix.

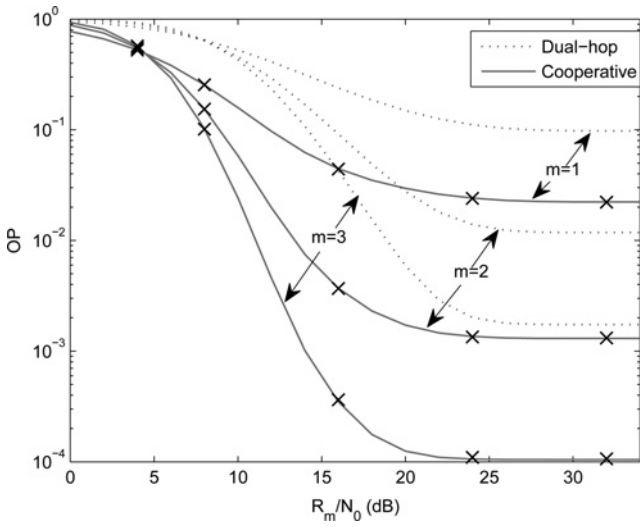

Fig. 2 *Outage probability against R_m/N_0*

The mark 'x' denotes simulated results whereas dash and solid lines stand for analytical ones

4 Illustrative results

This section provides results to validate the proposed outage probability expression, illustrate the performance behaviour, and compare the outage performance between underlay dual-hop cognitive networks in [9] and underlay cooperative cognitive networks considered in this paper. Without loss of generality, average fading powers are randomly selected as $\lambda_{03} = 1.5709$, $\lambda_{01} = 3.9528$, $\lambda_{13} = 31.6228$, $\lambda_{12} = 11.1803$, $\lambda_{02} = 1$. Also, we take the required transmission rate of $G = 1$ bps/Hz as an example. Moreover, we assume all m_{ab}'s to be identical, that is, $m_{ab} = m$ for any $\{a, b\}$, to limit case-studies.

The outage behaviour against R_m/N_0 is shown in Fig. 2 where 10^8 channel realisations are run to generate simulated results. In this figure, T_m/N_0 is fixed at 10 dB and the fading severity parameter m varies from 1 to 3. It is observed that simulation and analysis are in excellent agreement, justifying the validity of the derived expression in (3). In addition, several interesting insights into the outage performance of underlay cooperative cognitive networks are exposed from the analytical results as follows:

- For low-to-moderate values of R_m, the outage probability is inversely proportional to R_m. This comes from the fact that R_m upper-bounds the transmit power of unlicensed users and thus, the higher R_m results in the higher transmit power, eventually reducing the outage probability.
- At high values of R_m (e.g. about 25 dB), the performance is saturated. This is because at these values, the transmit power of unlicensed users is approximated as $\min((R_m/p_{a3}),\ T_m) \sim T_m$, causing the same outage for any increase in R_m.

$$f_{212a}^{a,b,c,d} = \int_0^k \frac{x^a (k-x)^j \Gamma(b,\ (\beta_{02}v(k-x) + \beta_{01}vk + \beta_{03})\mu)\Gamma(c,\ \beta_{13}\mu + \beta_{12}\chi x)}{(\beta_{02}v(k-x) + \beta_{01}vk + \beta_{03})^b (\beta_{13}+\beta_{12}vx)^d}\mathrm{d}x \tag{21}$$

$$f_{212b}^{a,b,c} = \int_0^k \frac{x^a (k-x)^j \Gamma(b,\ (\beta_{02}v(k-x) + \beta_{01}vk + \beta_{03})\mu)\mathrm{e}^{-\beta_{12}\chi x}}{(\beta_{13}+\beta_{12}vx)^c (\beta_{02}v(k-x) + \beta_{01}vk + \beta_{03})^b}\mathrm{d}x \tag{22}$$

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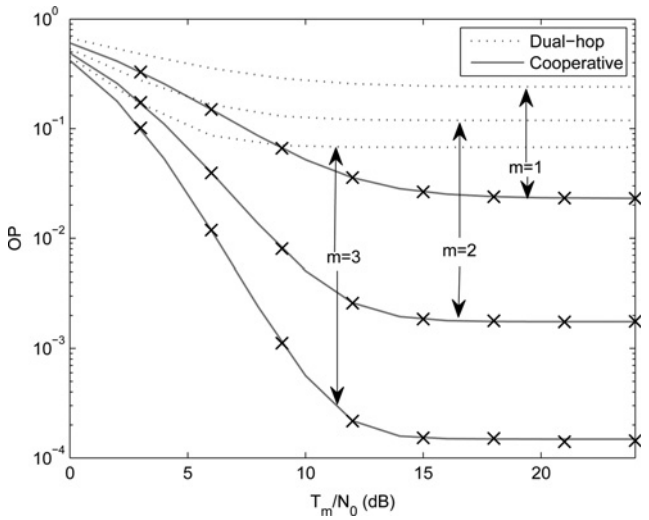

Fig. 3 *Outage probability against T_m/N_0*

The mark 'x' denotes simulated results whereas dash and solid lines stand for analytical ones

- In comparison with the dual-hop communications in [9], the cooperative one under consideration of this paper is dramatically better. This advantage comes at almost no cost of power and bandwidth, and thus, the cooperative communications in cognitive radio networks should be further exploited for performance improvement.
- All the results are reasonable in the sense that the performance of both communication protocols is significantly degraded with the increase in fading severity (i.e. smaller m).

Fig. 3 demonstrates the performance behaviour against T_m/N_0. The fading severity parameter m also changes from 1 to 3 and R_m/N_0 is constant at 15 dB. It is shown that simulated results perfectly agree analytical ones, verifying the accuracy of the proposed expression. Additionally, the results illustrate the performance saturation of underlay cooperative cognitive networks. We can explain this phenomena as follows. When T_m is over a threshold (e.g. about 15 dB in Fig. 3), the transmit power of unlicensed users is approximated as $\min\left(\left(R_m/p_{a3}\right),\ T_m\right) \sim \left(R_m/p_{a3}\right)$. That means, it is totally upper-bounded by R_m, leading to the same outage probability for any increase in T_m.

Similar to the results in Fig. 2, those in Fig. 3 expose that the cooperative communications considerably outperforms the dual-hop one and this superiority is dramatically increased with respect to the increase in m. This again shows the importance of taking the direct channel between the #0 user and the #2 user into account in cognitive radio networks.

5 Conclusion

The main contribution of this paper is the quite general outage probability expression of underlay cooperative cognitive networks over Nakagami-m fading channels. However, based on analytical results quickly generated by this expression for performance comparison between dual-hop and cooperative cognitive networks, and for behaviour illustration of underlay cognitive networks, several interesting discoveries and insights are exposed. More specifically, in cognitive radio networks the dual-hop communications is significantly inferior to the cooperative communications in a fair comparison in terms of power and bandwidth; and both communication protocols saturate at either high maximum interference power or high maximum transmit power.

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8 Appendix

This section is devoted to the derivation of (21), (22), (25)–(27) in Section 3.2. We start with $f_{221a}^{n,\beta;\Omega,\Lambda}$ in (25). Its closed-form depends on the values of n and β

- *Case 1:* $n \geq 0$ and $\beta < 0$

$$f_{221a}^{n,\beta;\Omega,\Lambda} = (-\beta)^{-(n+1)}\gamma\big(n+1,\ -\beta\Lambda\big) - (-\beta)^{-(n+1)}\gamma\big(n+1,\ -\beta\Omega\big) \tag{28}$$

This result comes from the definition of $\gamma(a, x)$.

- *Case 2:* $n \geq 0$ and $\beta \geq 0$

$$f_{221a}^{n,\beta;\Omega,\Lambda} = \mathrm{e}^{\beta\Lambda}\left(\frac{\Lambda^n}{\beta} + \sum_{k=1}^{n}(-1)^k\frac{n(n-1)...(n-k+1)}{\beta^{k+1}}\Lambda^{n-k}\right) - \mathrm{e}^{\beta\Omega}\left(\frac{\Omega^n}{\beta} + \sum_{k=1}^{n}(-1)^k\frac{n(n-1)...(n-k+1)}{\beta^{k+1}}\Omega^{n-k}\right) \tag{29}$$

This result is obtained with the help of [19, eq. (2.321.2)].

- *Case 3:* $n < 0$ and $\beta < 0$

$$\begin{aligned} f_{221a}^{n,\beta;\Omega,\Lambda} &= (-1)^{-n}\frac{(-\beta)^{-n-1}Ei(\beta\Omega)}{(-n-1)!} \\ &+ \frac{\mathrm{e}^{\beta\Omega}}{\Omega^{-n-1}}\sum_{k=0}^{-n-2}\frac{(-1)^k(-\beta)^k\Omega^k}{(-n-1)(-n-2)...(-n-1-k)} \\ &- (-1)^{-n}\frac{(-\beta)^{-n-1}Ei(\beta\Lambda)}{(-n-1)!} \\ &- \frac{\mathrm{e}^{\beta\Lambda}}{\Lambda^{-n-1}}\sum_{k=0}^{-n-2}\frac{(-1)^k(-\beta)^k\Lambda^k}{(-n-1)(-n-2)...(-n-1-k)} \end{aligned} \tag{30}$$

We solve the above with the help of [19, eq. (2.385.4)].

- *Case 4:* $n < 0$ and $\beta \geq 0$

$$\begin{aligned} f_{221a}^{n,\beta;\Omega,\Lambda} &= -\mathrm{e}^{\beta\Lambda}\sum_{k=1}^{-n-1}\frac{\beta^{k-1}}{(-n-1)\cdots(-n-k)\Lambda^{-n-k}} \\ &+ \frac{\beta^{-n-1}}{(-n-1)!}Ei(\beta\Lambda) \\ &+ \mathrm{e}^{\beta\Omega}\sum_{k=1}^{-n-1}\frac{\beta^{k-1}}{(-n-1)\cdots(-n-k)\Omega^{-n-k}} \\ &- \frac{\beta^{-n-1}}{(-n-1)!}Ei(\beta\Omega) \end{aligned} \tag{31}$$

This result is obtained with the help of [19, eq. (2.324.2)].

In (30) and (31), $Ei(x) = -\int_{-x}\frac{\mathrm{e}^{-t}}{t}\mathrm{d}t$ is the exponential integral function in [19, eq. (8.211)]. This is a built-in function in several computation softwares (e.g. Matlab).

The integral $f_{221b}^{a,b}$ in (26) is computed by changing the integral variable and using the binomial expansion

$$f_{221b}^{a,b} = \frac{\mathrm{e}^{(((\beta_{12}-\beta_{02})\beta_{13}\chi)/\beta_{12}\nu)}}{\big(\beta_{12}\nu\big)^{a+1}} \times \sum_{p=0}^{a}\binom{a}{p}\big(-\beta_{13}\big)^{a-p}f_{221a}^{p-b,\ (((\beta_{02}-\beta_{12})\chi)/\beta_{12}\nu);\beta_{13},\ \beta_{13}+\beta_{12}\nu k} \tag{32}$$

The integral $f_{221c}^{a,b,c}$ in (27) is solved in the closed form using the expansion in (16)

$$\begin{aligned} f_{221c}^{a,b,c} &= \int_0^k \frac{x^a\left(\Gamma(b)\mathrm{e}^{-(\beta_{13}\mu+\beta_{12}\chi x)}\sum_{s=0}^{b-1}\left(\left(\big(\beta_{13}\mu+\beta_{12}\chi x\big)^s\right)/s!\right)\right)\mathrm{e}^{\beta_{02}\chi x}}{\big(\beta_{13}+\beta_{12}\nu x\big)^c}\mathrm{d}x \\ &= \Gamma(b)\mathrm{e}^{-\beta_{13}\mu}\sum_{s=0}^{b-1}\frac{\mu^s}{s!}f_{221b}^{a,c-s} \end{aligned} \tag{33}$$

Before deriving $f_{212a}^{a,b,c,d}$ in (21) and $f_{212b}^{a,b,c}$ in (22), we solve the following integral

$$f_G^{a,j,b,c} = \int_0^k \frac{x^a(k-x)^j\mathrm{e}^{(\beta_{02}-\beta_{12})\chi x}}{\big(\beta_{02}\nu(k-x)+\beta_{01}\nu k+\beta_{03}\big)^b\big(\beta_{13}+\beta_{12}\nu x\big)^c}\mathrm{d}x \tag{34}$$

Applying the binomial expansion and changing the integral variable repeatedly, $f_G^{a,j,b,c}$ is reduced to (see (35))

The computation of $f_G^{a,j,b,c}$ is completed in consideration of two cases

$$\begin{aligned} f_G^{a,j,b,c} &= \sum_{l=0}^{j}\sum_{z=0}^{a+l}\binom{j}{l}\binom{a+l}{z}\frac{(-1)^{a-z}k^{j-l}\beta_{13}^{a+l-z}\mathrm{e}^{(((\beta_{12}-\beta_{02})\beta_{13}\mu)/\beta_{12})}}{\big(\beta_{12}\nu\big)^{a+l+1}} \\ &\times \int_{\beta_{13}}^{\beta_{13}+\beta_{12}\nu k}\frac{y^z\mathrm{e}^{(((\beta_{02}-\beta_{12})\mu y)/\beta_{12})}}{\big(\beta_{02}\nu k+\beta_{01}\nu k+\beta_{03}+\big(\big(\beta_{02}\beta_{13}\big)/\beta_{12}\big)-\big(\beta_{02}/\beta_{12}\big)y\big)^b y^c}\mathrm{d}y \end{aligned} \tag{35}$$

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• *Case 1:* $z \geq c$. Applying the binomial expansion and changing the integral variable one more time, we obtain the closed form of $f_G^{a,j,b,c}$ as (see (36))

• *Case 2:* $z < c$.Applying the partial fraction expansion method, $f_G^{a,j,b,c}$ is solved in the closed form as (see (37))

where constants A_m and B_n are defined as

$$A_m = \frac{(-1)^{m-1}}{(m-1)!} \frac{\prod_{q=0}^{m-2}(c-z+q)}{\delta^{c-z+m-1}} \tag{38}$$

$$B_n = \frac{(-1)^{b}}{(n-1)!} \frac{\prod_{q=0}^{n-2}(b+q)}{\delta^{b+n-1}} \tag{39}$$

with

$$\delta = \frac{(\beta_{02}vk + \beta_{01}vk + \beta_{03})\beta_{12}}{\beta_{02}} + \beta_{13} \tag{40}$$

Now we return to $f_{212a}^{a,b,c,d}$. Using (16) results in the closed form of (see (41))

Similarly, using (16), we can express $f_{212b}^{a,b,c}$ in the closed form as (see (42))

$$f_G^{a,j,b,c} = \sum_{l=0}^{j}\sum_{z=0}^{a+l}\sum_{g=0}^{z-c}\binom{j}{l}\binom{a+l}{z}\binom{z-c}{g}\frac{(-1)^{a+g-z+1}\beta_{02}^{c-z-1}k^{j-l}\beta_{13}^{a+l-z}\mathrm{e}^{(((\beta_{02}-\beta_{12})(\beta_{02}vk+\beta_{01}vk+\beta_{03})\mu)/\beta_{02})}}{\beta_{12}^{a+l-g}v^{a+l+1}\left(\beta_{12}\{\beta_{02}vk+\beta_{01}vk+\beta_{03}\}+\beta_{02}\beta_{13}\right)^{c+g-z}} \times f_{221a}^{g-b,(((\beta_{12}-\beta_{02})\mu)/\beta_{02});\beta_{02}vk+\beta_{01}vk+\beta_{03},\beta_{01}vk+\beta_{03}} \tag{36}$$

$$f_G^{a,j,b,c} = \sum_{l=0}^{j}\sum_{z=0}^{a+l}\binom{j}{l}\binom{a+l}{z}\frac{(-1)^{a-b-z}k^{j-l}\beta_{13}^{a+l-z}\mathrm{e}^{(((\beta_{12}-\beta_{02})\beta_{13}\mu)/\beta_{12})}}{\beta_{02}^{b}\beta_{12}^{a+l+1-b}v^{a+l+1}} \times \left(\begin{array}{l}\dfrac{\sum_{m=1}^{b}A_m f_{221a}^{m-b-1,(((\beta_{02}-\beta_{12})\mu)/\beta_{12});-(((\beta_{02}vk+\beta_{01}vk+\beta_{03})\beta_{12})/\beta_{02}),-(((\beta_{01}vk+\beta_{03})\beta_{12})/\beta_{02})}}{\mathrm{e}^{(\beta_{12}-\beta_{02})(((\beta_{02}vk+\beta_{01}vk+\beta_{03})/\beta_{02})+(\beta_{13}/\beta_{12}))\mu}} \\ +\sum_{n=1}^{c-z}B_n f_{221a}^{z+n-c-1,((\beta_{02}-\beta_{12})\mu/\beta_{12});\beta_{13},\beta_{13}+\beta_{12}vk}\end{array}\right) \tag{37}$$

$$\begin{aligned} f_{212a}^{a,b,c,d} &= \int_0^k \frac{\left(\Gamma(b)\mathrm{e}^{-(\beta_{02}v(k-x)+\beta_{01}vk+\beta_{03})\mu}\sum_{w=0}^{b-1}\left(\left[\left(\beta_{02}v(k-x)+\beta_{01}vk+\beta_{03}\right)\mu\right]^{w}\right)/w!\right)}{x^{-a}(k-x)^{-j}\left(\beta_{02}v(k-x)+\beta_{01}vk+\beta_{03}\right)^{b}\left(\beta_{13}+\beta_{12}vx\right)^{d}} \\ &\quad\times\left(\Gamma(c)\mathrm{e}^{-(\beta_{13}\mu+\beta_{12}\chi x)}\sum_{u=0}^{c-1}\frac{\left(\beta_{13}\mu+\beta_{12}\chi x\right)^{u}}{u!}\right)\mathrm{d}x \\ &= \frac{\Gamma(b)\Gamma(c)}{\mathrm{e}^{(\{\beta_{02}+\beta_{01}\}vk+\beta_{03}+\beta_{13})\mu}}\sum_{w=0}^{b-1}\sum_{u=0}^{c-1}\frac{\mu^{w+u}}{w!u!}f_G^{a,j,b-w,d-u} \end{aligned} \tag{41}$$

$$\begin{aligned} f_{212b}^{a,b,c} &= \int_0^k \frac{x^a(k-x)^j\mathrm{e}^{-\beta_{12}\chi x}}{\left(\beta_{13}+\beta_{12}vx\right)^{c}\left(\beta_{02}v(k-x)+\beta_{01}vk+\beta_{03}\right)^{b}}\left(\Gamma(b)\mathrm{e}^{-(\beta_{02}v(k-x)+\beta_{01}vk+\beta_{03})\mu}\sum_{w=0}^{b-1}\frac{\left[\left(\beta_{02}v(k-x)+\beta_{01}vk+\beta_{03}\right)\mu\right]^{w}}{w!}\right)\mathrm{d}x \\ &= \Gamma(b)\mathrm{e}^{-(\{\beta_{02}+\beta_{01}\}vk+\beta_{03})\mu}\sum_{w=0}^{b-1}\frac{\mu^{w}}{w!}f_G^{a,j,b-w,c} \end{aligned} \tag{42}$$

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